

eCommerce VIEW Assignment (MySQL)

Business Scenario

An eCommerce system has:

- Customers
- Products
- Orders
- Order Items

1. Database Schema (Setup)

```
-- =====  
  
-- CREATE DATABASE  
  
-- =====  
  
CREATE DATABASE ecommerce_view_demo;  
USE ecommerce_view_demo;  
  
  
-- =====  
  
-- TABLES  
  
-- =====  
  
  
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY AUTO_INCREMENT,  
    customer_name VARCHAR(100),  
    email VARCHAR(100),  
    city VARCHAR(100)  
);  
  
  
CREATE TABLE products (  
    product_id INT PRIMARY KEY AUTO_INCREMENT,  
    product_name VARCHAR(100),
```

```
category VARCHAR(50),  
price DECIMAL(10,2)  
);
```

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY AUTO_INCREMENT,  
    customer_id INT,  
    order_date DATE,  
    status VARCHAR(50),  
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)  
);
```

```
CREATE TABLE order_items (  
    order_item_id INT PRIMARY KEY AUTO_INCREMENT,  
    order_id INT,  
    product_id INT,  
    quantity INT,  
    FOREIGN KEY (order_id) REFERENCES orders(order_id),  
    FOREIGN KEY (product_id) REFERENCES products(product_id)  
);
```

```
-- =====  
-- SAMPLE DATA  
-- =====
```

```
INSERT INTO customers (customer_name, email, city) VALUES  
( 'Aung Aung', 'aung@gmail.com', 'Yangon'),  
( 'Su Su', 'susu@gmail.com', 'Mandalay'),  
( 'Kyaw Kyaw', 'kyaw@gmail.com', 'Yangon');
```

```
INSERT INTO products (product_name, category, price) VALUES
('Laptop', 'Electronics', 1500000),
('Phone', 'Electronics', 800000),
('Shoes', 'Fashion', 120000);
```

```
INSERT INTO orders (customer_id, order_date, status) VALUES
(1, '2024-01-10', 'Completed'),
(2, '2024-01-11', 'Pending'),
(1, '2024-01-12', 'Completed');
```

```
INSERT INTO order_items (order_id, product_id, quantity) VALUES
(1, 1, 1),
(1, 2, 2),
(2, 3, 1),
(3, 2, 1);
```

Assignment Questions & Answers

 **Q1. Create a VIEW to show customer basic information (hide email)**

 **Q2. Create a VIEW to show all completed orders**

 **Q3. Create a JOIN VIEW for order details (customer + product)**

 **Q4. Create a VIEW to calculate total sales per order**

 **Q5. Create a VIEW for high-value orders (above 1,000,000)**

✓ Q6. Create a VIEW with WITH CHECK OPTION (only completed orders allowed)

✓ Q7. Try to update data using VIEW (Updatable View)

✗ Q8. Try invalid update (should fail)

✓ Q9. Create a VIEW for customer purchase summary

✓ Q10. Create a VIEW using ALGORITHM = TEMPTABLE

✓ Q11. Create a VIEW for security (hide price)

✓ Q12. Show all views in database

✓ Q13. Show view definition

✓ Q14. Drop a view

Additional Questions

1. Create a view for **pending orders only**
2. Create a view showing **top 3 expensive products**
3. Create a view showing **customer orders by city**
4. Create a view for **monthly sales report**
5. Create a view showing **products not ordered**